Mohammad Ful Hossain Seikh

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Professional Objective

To develop expertise in experimental Astroparticle Physics through cutting-edge research at world-class laboratories, while contributing to the advancement of ultra-high energy neutrino detection technologies.

Education

Ph.D. in Physics *2020–Present*
University of Kansas, USA **3.98/4.0**

M.S. in Electrical Engineering *2024–2025*
University of Kansas, USA **3.87/4.0**

M.Sc. in Physics *2018–2020*
Aligarh Muslim University, India **9.170/10** (Gold Medalist)

B.Sc. Honors in Physics *2015–2018*
Aligarh Muslim University, India **9.186/10** (Gold Medalist)

Higher Secondary (12th) *2014*
Central Board of Secondary Education, India **9.120/10** (First Position)

Secondary (10th) *2012*
Central Board of Secondary Education, India **10/10** (Certificate of Merit)

Research Experience

Primary Analyzer, Askaryan Radio Array (ARA) Station 1 *2021–Present*
Conducting data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection with Prof. D. Besson at University of Kansas (KU).

EECS Master's Project Research *2024–2025*
Developed **AAFIYA**: **A**ntenna **A**nalysis in **F**requency-domain for **I**mpedance and **Y**ield **A**ssessment, a modular Python toolkit for rigorous, reproducible analysis and visualization of antenna S-parameters, impedance, and gain/beam patterns from both measurement and simulation data, (Report), (Slides).

IceCube Neutrino Observatory Monitoring *2021–2025*
Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.

Physics Master's Thesis Research *2019–2020*
Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report "Light Sterile Neutrinos: Status and Prospects" (DOI:10.13140/RG.2.2.13777.66405).

Summer Research Program *May–Jul 2019*
Studied the role of θ_{13} mixing angle in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at Institute of Physics, India, producing technical report "Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm" (DOI: 10.13140/RG.2.2.34421.12009).

Winter Internship Program *Dec 2019–Jan 2020*
Researched "Weak Interaction Physics" at IIT-BHU with Dr. G. Abbas.

Physics Undergraduate Project Research *2017–2018*
"Quarks & Leptons": Detailed study of Standard Model interactions with Prof. M.S. Athar at AMU [View Handwritten Report].

Field Experience

NSF's Summit Station, Greenland Field Campaign *Apr–May 2025*
Participated in the Radio Neutrino Observatory – Greenland (RNO-G) calibration campaign, performing in-situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications

First-Author Works

- **Seikh, M.F.H.** *(2025).* "A geometrical approach to neutrino oscillation parameters". *AIP Advances 1 October 2025; 15 (10): 105204.* DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** *2025.* "First Radio Neutrino & Cosmic Rays Astronomy Workshop in India". *European Physics Journal-Special Topics (EPJ-ST).* arXiv:2509.20704 DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** *2025.* "Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array". PoS(ICRC2025)1171. arXiv:2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** *2025.* "Observation of Radio Solar Flares in RNO-G". PoS(ICRC2025)1365.
- **Seikh, M.F.H.** *(2024).* "Physics of radio antennas". *European Physics Journal-Special Topics (EPJ-ST).* arXiv:2411.07507 DOI: 10.1140/epjs/s11734-025-01500-7
- **Seikh, M.F.H. for the ARA Collaboration** *(2024).* "Askaryan Radio Array: searching for the highest energy neutrinos". *European Physics Journal-Special Topics (EPJ-ST).* arXiv:2411.01761 DOI: 10.1140/epjs/s11734-025-01708-7
- **Seikh, M.F.H. & Besson, D.** *(2024).* "IceCube Neutrino Observatory: A Cold Discovery Hub". *Page 14-23, Physics Bulletin, Physics Department, AMU*
- **Seikh, M.F.H. for the ARA Collaboration** *(2023).* "Calibration and Physics with ARA Station 1". *PoS(ICRC2023)1163.* arXiv:2308.07292
- **Seikh, M.F.H.** *(2020).* "Neutrinos and the 40 Future Routes". *Physics Bulletin*, Physics Department, AMU. DOI: 10.13140/RG.2.2.27981.10729

Collaborative Works

Full publication (includes contributions to ARA, RNO-G, ARIANNA, IceCube, IceCube Gen2, PUEO, TAROGE-M and RET collaborations) record available on:

- Inspire HEP
- Google Scholar
- ResearchGate
- ORCiD
- Scopus

Professional Service

Editorial

- **Guest Editor** for Springer's EPJ-ST journal: "Radio Detection of Ultra-high Energy Neutrino & Cosmic Rays" (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko *(2024–2025)*
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU's Society of Physics Students [View Issue] *(June 2022)*

Research Article Review

- **Peer Reviewer:** for World Scientific's IJMP-A journal (1 article, 1 review) (2025)
- **Peer Reviewer:** for Springer's EPJ-ST journal (5 articles, 11 reviews) *(2024–2025)*
- **Internal ARA Collaboration Reviewer:** Pawan Giri for ARA Collaboration, "A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array", 32nd International Symposium on Lepton Photon Interactions at High Energies *(November 2025)*

Professional Leadership/Membership

- **American Physical Society (APS)**: Graduate Student Member *(2020–Present)*
- **National Society of Black Physicists**: Student Member *(2021–Present)*
- **Physics & Astronomy Graduate Student Organization (GSO), KU**:
  - ✓ Graduate Students Representative - Member at Large *(2025–Present)*
  - ✓ Graduate Students Representative - Liaison *(2024–2025)*
  - ✓ GSO Executive Officer *(2023–2024)*
- **Sir Syed Global Scholar Award (SSGSA)**:
  - ✓ Peer Mentor *(2023–Present)*
  - ✓ Team Lead, Promotions & Outreach *(2023–2025)*
  - ✓ Member, Mentorship & Applications Team *(2021–2023)*
  - ✓ AMU Campus Team Member *(2019–2020)*
- **Society of Physics Students, KU Chapter**: Graduate President *(2021–2022)*
- **Sigma Pi Sigma (National Physics Honor Society)**: Lifetime Member
- **AMU Society of Physics**: Member *(2018–Present)*

Technical Proficiencies

- **Programming Languages**: C/C++, Python, FORTRAN
- **Physics Software**: CERN ROOT, AraRoot, Wolfram Mathematica, HFSS
- **Operating Systems**: Linux (Ubuntu), Windows, macOS
- **Document Preparation**: LaTeX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals**:
  - ✓ M.Sc. Physics (Top among 50 students), AMU *(2020)*
  - ✓ B.Sc. Honors Physics (Top among 120 students), AMU *(2018)*
  - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU *(2018)*

- **Abdul Aziz Medal** for B.Sc. Physics, AMU *(2018)*
- **Undergraduate Topper Scholarship**, UGC India *(2020)*
- **University Merit Financial Award**, AMU *(2019–2020)*
- **First Position** in Higher Secondary (12th) among 40 students *(2014)*
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th) *(2012)*

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics *(2025–2026)*
- **McKeithan Summer Fellowship**, KU Physics *(2025)*
- **Graduate Student Award for Distinguished Service**, KU *(2024–2025)*
- **Wallace S. Strobel Scholarship**, KU EECS *(2024–2025)*
- **Ralston Summer Fellowship**, KU Physics *(2024)*
- **Doctoral Student Research Fund**, KU *(2024)*
- **Distinguished Students Award**, APS FIP *(2021–2022)*
- **Carl & Catherine Chaffee Scholarship**, KU Physics *(2020–2021)*

Conference Travel Awards

- **Graduate Student Travel Fund**, KU Physics *(2023, 2025)*
- **DAP Travel Grant**, APS *(2024)*
- **DPF Travel Award**, APS *(2022, 2024)*
- **CLAS Graduate Scholarly Development Fund**, KU *(2022)*

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association *(2015–2019)*
- **Sir Syed Global Scholar Award** for prospective graduate students *(2018)*

Invited Lectures, Seminars, & Colloquia

- **Radio Engineering for Particle Astrophysics Experiments**
  HEP Seminar, University of Nebraska-Lincoln *(March 26, 2025)*
- **Radio Engineering for Particle Astrophysics Experiments**
  Astroparticle Seminar, Ohio State University *(January 22, 2025)*
- **Radio Engineering & Antenna Physics**
  Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage *(July 2–5, 2024)*
- **UHE Neutrino & Cosmic Ray Detectors**
  ICEHAP Seminar, Chiba University, Japan *(August 9, 2023)*
- **Antenna Physics & Measurements**
  Lecture Series, IEI Summer School 2023, South Dakota School of Mines *(July 2–5, 2023)*
- **UHE Neutrino & Cosmic Ray Detectors**
  Physics Colloquium, AMU, India *(January 12, 2023)*
- **Antennas in UHE Neutrino Detectors**
  Particle Physics Seminar, University of Delaware *(October 4, 2022)*

Outreach & Leadership

- Co-organized and hosted India's inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU *(April 2024)*

- Conducted outreach seminar on **Physics Research Opportunities** at AMU *(2021)*
- Participated in India's first **Science Leadership Workshop** *(2020)*
- Volunteer for **One Person One Tree** plantation drive *(2017)*
- Participant in **Blood Donation Camp** *(2017)*

Conference Presentations

Oral Presentations

- **RNO-G in-ice Antennas**
  RNO-G Collaboration Meeting, Uppsala University *(September 9–14, 2024)*
- **Radio Frequency Instrumentation**
  Radio Neutrino Workshop, AMU, India *(April 18–20, 2024)*
- **Towards Five Station ARA Analysis**
  APS April Meeting, Sacramento *(April 3–6, 2024)*
- **ARA Antenna Models**
  ARA Collaboration Meeting, Ohio State University *(March 5–8, 2024)*
- **Current Status & Future of ARA**
  APS Prairie Section Meeting, University of Missouri *(November 30–December 2, 2023)*
- **Introduction to Antenna Physics**
  RET Collaboration Meeting, Vrije University Brussels *(November 8–11, 2023)*
- **Gain Calibration of RNO-G Antennas**
  RNO-G Fall Meeting, Penn State University *(September 11–15, 2023)*
- **Galactic Noise in Radio Neutrino Experiments**
  Astro Seminar, KU *(October 28, 2022)*
- **Towards ARA Station 1 Calibration**
  APS April Meeting, New York City *(April 9–12, 2022)*
- **RNO-G Ice Calibration Planning**
  RNO-G Collaboration Meeting, University of Chicago *(April 8, 2022)*

Poster Presentations

- **CW Identification & Filtering Techniques in ARA**
  39th ICRC, Geneva, Switzerland *(July 19, 2025)*
- **Observation of radio solar flares in RNO-G**
  39th ICRC, Geneva, Switzerland *(July 15, 2025)*
- **Gain Measurements of LPDA Antennas**
  APS April Meeting, Sacramento *(April 3–6, 2024)*
- **Overview of ARA Detector: Five Station Analysis**
  ICTS Neutrino School, India *(April 22–May 3, 2024)*
- **Calibration of ARA Station 1**
  38th ICRC, Nagoya University, Japan *(July 29, 2023)*

Professional Development

- **Responsible Conduct of Research**, (CITI Program, *2021, 2025*) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, *2020*) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, *2019*) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad) - Distinction